

LEXIPILOT

Private Adaptive Vocabulary Learning Agent
on AMD Radeon

Model-selected read-only tools. Validated structured plans. Deterministic progress writes.

TRACK 2

- Qwen3-8B planning
- vLLM + ROCm
- Dedicated AMD Radeon inference
- Persistent learner memory
- Safe deterministic controller

<PARTICIPANT_OR_TEAM_NAME>

github.com/appleboyecat/LexiPilot

Static vocabulary tools miss the learner

Static lists

The same sequence is shown regardless of due dates, repeated mistakes, or available time.

Fragmented memory

Review history exists, but the learner still decides manually what matters today.

Weak adaptation

A wrong answer rarely changes the next activity or generated practice.

Unsafe LLM control

Unrestricted model writes can fabricate words, corrupt progress, or expose private context.

A hybrid Agent: adaptive decisions, controlled actions



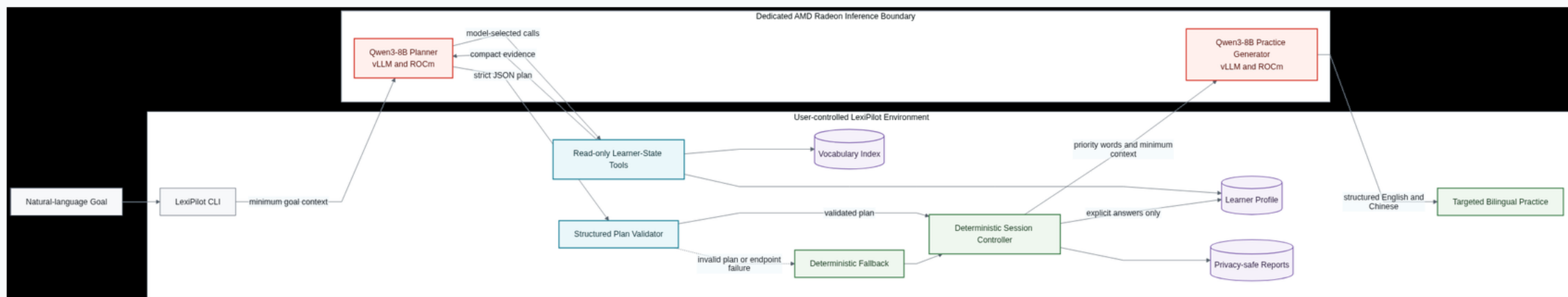
Model responsibility

Inspect learner state, choose read-only tools, and propose a plan grounded in returned evidence.

Controller responsibility

Present cards, validate input, record explicit answers, apply spaced repetition, and finalize exactly once.

Hybrid Agent architecture and trust boundaries



Read-only planning

The model cannot record answers or save progress.

Validated execution

Unknown words and malformed plans are rejected.

Minimum disclosure

Only task-relevant context crosses the endpoint boundary.

Safety is enforced in code, not promised by the model

Read-only planning tools

No record_answer or save_session_summary schema is exposed.

Strict plan validation

Exact schema, real tool evidence, known words, and practical limits.

Deterministic writes

Only explicit y/n answers update spaced repetition.

Failure containment

Timeouts, fake Tool Calling, bad JSON, and unknown tools fall back.

Privacy-safe reporting

No keys, private URLs, prompts, responses, or full profiles.

Qwen3-8B on a dedicated Radeon inference endpoint

LexiPilot client

OpenAI-compatible non-streaming requests

vLLM server

Hermes tool parser and automatic Tool Calling

Qwen3-8B

Planning and bilingual practice generation

ROCm + AMD Radeon

Dedicated Cloud GPU instance

Deployment Evidence

- GPU model: Capture during final Radeon demo
- ROCm version: Capture during final Radeon demo
- vLLM version: Capture during final Radeon demo
- GPU activity: Capture during final Radeon demo
- Endpoint Tool Calling: verified by script

Learner state stays in the controlled LexiPilot environment; minimum task context is sent to the dedicated endpoint.

Measured thinking-mode comparison

1 warm-up + 5 measured requests per mode and workload | temperature 0 | max tokens 700

Workload / Metric	Thinking enabled	Thinking disabled	Observed result
Planning median latency	13.7371 s	13.7773 s	No clear improvement
Planning Tool Calling	100%	100%	Same reliability
Generation median latency	7.0341 s	6.5433 s	6.98% lower median
Generation validation	100%	100%	Same reliability
Generation completion tokens/s	17.9127	19.2563	7.50% higher

Measurement scope Client-observed end-to-end values include network, endpoint, scheduling, and serving overhead - not raw GPU throughput.

A reproducible Agent pattern for private adaptive learning

Submission impact

A useful learning workflow with real persistent memory, safe model autonomy, deterministic writes, and Radeon-hosted inference.

Reproducible by judges

A sanitized 40-word index, synthetic profile, model-free smoke test, endpoint verifier, and benchmark script.

Roadmap

- Localhost AMD Radeon workstation deployment
- Optional GUI after the CLI trust model remains stable
- Multilingual practice beyond English and Chinese
- Richer long-term learner analytics

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